1. **What is C language?**

C is a programming language developed at AT & T's Bell Laboratories of USA in 1972.

The C programming language is a standardized programming language developed in theearly 1970s by Ken Thompson and Dennis Ritchie for use on the UNIX operatingsystem.

2) What is an algorithm?

Ans: An algorithm is a step-by-step method of performing any task.

3) What is a flow chart?

Ans: A flowchart is a type of diagram that represents an algorithm or process, showingthe steps as boxes of various kinds, and their order by connecting these with arrows.

4) What is a C Preprocessor?

Ans: C Preprocessor is a program that processes our source program before it is passed tothe compiler.

5) What is the use of header files as used in C programming?

Ans: Header files are used to have declarations. It is simple to include a single header filethan writing allthe needed functions prototypes.

6) What is the Structure of a C Program?

Ans: Documentation Section, Linking Section, Definition Section, Global declarationSection, main function, subprogram section.

7) Explain some Header Files?

Ans:

stdio.h

This is standard input/output header file inwhich Input/Outputfunctions are declared2

conio.h

This is console input/output header file3

8. What is the use of main() function?

Ans: main() is the starting point of program execution.

9) What are the types of constants in c?

Ans: C constants can ba divided into two categories:1)

Primary constants (Numerical)2)

Secondary constants (Character)

10) What is a Compiler?

Ans: Compiler is a computer program (or set of programs) that transforms sourcecode written in a programming language (the source language) into another computerlanguage (the target language, often having a binary form known as object code).

11) What is a Translator?

Ans: A translator is a computer program that translates a program written in agiven programming language into a functionally equivalent program in a differentlanguage.

12) What is a Interpreter?

Ans: An Interpreter is a computer program that directly executes, i.e. performs ,instructions written in a programming or scripting language, without previously batch-compiling them into machine language.

13) What is a Token in C?

Ans: A Token is the basic building block of a **C. (or)** the basic element recognized bythe compiler is the " **Token** ."C Tokens are:Key Words, Identifier, Constants, String – literal, Operator, Punctuators

14) What are Printf() and scanf() Functions :

Ans: printf() and scanf() functions are inbuilt library functions in C which are availablein which are available in " stdio.h" header file.



printf() function is used to print the "character, string, float, integer, octal andhexadecimal values" onto the output screen.

scanf() function is used to read character, string, numeric data from keyboard

15. What is a Data Type and List the different Data types?

Ans: C data types are defined as the data storage format that a variable can store a data toperform a specific operation.List of Data Types: 1)

Basic Data Types: Int, Float, Char, Double, long int2)

Enumeration Data Type: enum3)

Derived Data Type: Pointer, array, structure, union4)

Void Data Type: void

16) What is Integer Data Type?

Ans:

Integer data type allows a variable to store numeric values.

The storage size of int data type is 2 or 4 or 8 byte.

int (2 byte) can store values from -32,768 to +32,767

17. What is enum Data Type?

Ans:

Enumeration data type consists of named integer constants as a list.

It start with 0 (zero) by default and value is incremented by 1 for the sequentialidentifiers in the list.

Syntax:

enum identifier { enumerator-list };

18) What is a Void Data Type?

Ans: Void is an empty data type that has no value.

19) What is a comment in C?

Ans: Comments are like helping text in your C program and they are ignored by thecompiler. We can write in between /* and */ or // (Line Comments)

20) What is an Identifier in C?

Ans: A C identifier is a name used to identify a variable, function, or any other user-defined item. An identifier starts with a letter A to Z or a to z or an underscore _ followedby zero or more letters, underscores, and digits (0 to 9). C does not allow punctuationcharacters such as @, $, and % within identifiers. C is a

case sensitive

programminglanguage.

21) What is a Key word in C?

Ans: Keywords are reserved words in C and Keywords are may not be used as constantor variable or any other identifier names.

22) How many Keywords are there in C and List out the Keywords?

Ans : There are 33 reserved keywords are there in C. They are:auto else long switchbreak enum register typedefcase extern return unionchar float short unsignedconst for signed voidcontinue goto sizeof volatiledefault if static whiledo int struct _Packeddouble

23. Define a Variable?

Ans: A variable is nothing but a name given to a storage area that our programs canmanipulate. Each variable in C has a specific type, which determines the size and layoutof the variable's memory.

Syntax for variable Declaration:

DataType VariableList;int i,j;cahr c, ch;

30. What are Assignment Operators?

Ans:

Values for the variables are assigned using assignment operators. There are twotypes assignment operators are there:

1) Simple Assignment

(=)

2) Compound Assignment

(+= , -=, *=, %=)

31) What are Relational Operators?

Ans: Relational operators are used to find the relation between two variables. I.e. tocompare the values of two variables in a C program.. (Like >, <, >=, <=, !=, ==)

32)What are Logical Operators?

Ans:

These operators are used to perform logical operations on the given expressions.(Like &&, ||, !)

33)What are Bitwise Operators?

Ans:

These operators are used to perform bit operations. Decimal values are convertedinto binary values which are the sequence of bits and bit wise operators work on thesebits. (Like &, |, ~, ^, <<(Left Shift), >> (Right Shift))

34)What are Conditional (Ternary) Operators?

Ans:

Conditional operators return one value if condition is true and returns another valueis condition is false. (Like: ? and :)

35 What are Increment / Decrement Operators?

Ans:

Increment operators are used to increase the value of the variable by one anddecrement operators are used to decrease the value of the variable by one in C programs.(Like ++, --)

36)What are Special Operators?

Ans:

&(Address Operator), * (Pointer Operator), and Sizeof()

37)

What is sizeof operator?Ans:

It returns the number of bytes the operand occupies.

38)

What is pre-increment or post-increment?Ans:

++n (pre increment) increments n before its value is used in an assignmentoperation or any expression containing it. n++ (post increment) does increment after thevalue of n is used.

39)

What is type casting?Ans:

Converting a variable of one type to another type.

40)

What are the Format Specifiers or Type Specifiers or Conversion Specifiers?Ans:

%d (Integer), %f (Float), %c (Character), %l (Long Integer), %s (Strings), %u(Address with decimal value), %p (Address with Hexa Decimal Value in Small Letters),%x ((Address with Hexa Decimal Value in Capital Letters)

41)

What is a Statement in C?Ans:

A statement is a block of code that does something.

42)

Different Types of Statements?Ans:

Null Statement, Expression Statement, Compound Statement, Return Statement,Conditional Statements, Iterative or Looping Statements, Unconditional Statements

43)

Define Null Statement?Ans:

A "null statement" is a statement containing only a semicolon;

44)

Define Expression Statement?Ans:

When an expression statement is executed, the expression is evaluated according tothe rules outlined inExpressions and Assignments.

45)

Define Compound Statement?Ans:

A compound statement (also called a "block") typically appears as the body ofanother statement which is in between { and

Define Return Statement?Ans:

The

return

statement terminates the execution of a function and returns control tothe calling function. A

return

statement can also return a value to the calling function.

47)

Define Conditional Statements and give the list of them?Ans:

Conditional Statements which allows to perform actions depending upon someconditions provided by the programmer. The Different types of conditional statementsare:

1)

If Statement

2)

If else statement

3)

Nested- if else statement

4)

Switch Statement

48)

Write the Syntax for IF Statement?Ans

: if (<expression>)<statement>

49)

Write the Syntax for IF- ELSE Statement?Ans

: if (<expression>)<statement 1>else<statement 2>

50)

Write the Syntax for NESTED- IF - ELSE Statement?Ans

: if (<expression1>){if(<expression2>){<Statements>}else{<Statements>}}else<statement>

51)

Write the Syntax for Switch Case Statement?Ans

: switch (<expression>)(<case list> }

What are Iterative or Looping Statements?Ans

:

Iterative or Looping statement which executes the statements with in the compoundstatement by checking the condition, and performs same set of statements as a iterativeprocess or as a loop until the condition false.

The Iterative or Looping Statements are

:

While, do-while, for

53)

Write Syntax for WHILE Statement?Ans:

while (<expression>)<statement>

54)

Write Syntax for DO- WHILE Statement?Ans:

do <statement>while (<expression>);

55)

Write Syntax for FOR Statement?Ans

: for (<Initialization>;<Condition>;<Increment/Decrement>)<statement>

56)

What is the difference between for loop and while loop?Ans:

For Loop is used to execute a set of statements in fixed number of times. We useWhile loop when the number of iterations to be performed is not known in advance weuse while loop.

57)

What are Unconditional Statements?Ans

: goto and labeled statements, break Statement continue Statement.

58)

Define goto and labeled Statement, Write syntax for goto and labeled statements?Ans:

The

goto

statement transfers control to a label. The given label must reside in thesame function and can appear before only one statement in the same function.

Syntax:

goto <label>;<label>: <Statement>;

efine Break Statement, Write syntax for Break statement?Ans:

The

break

statement terminates the execution of the nearestenclosing

do

,

for

switch

, or

while

statement in which it appears. Control passes to thestatement that follows the terminated statement.

Syntax:

break;

60)

Define Continue Statement, Write syntax for Continue statement?Ans:

The

continue

statement passes control to the next iteration of the nearestenclosing

do

,

for

, or

while

statement in which it appears

Syntax:

continue;

61)

Define Type Qualifiers, and list them?Ans:

The keywords which are used to modify the properties of a variable are called typequalifiers. There are two types of qualifiers available in C language. They are,1) const 2)volatile

62)

Define const Keyword with Syntax?Ans:

Constants are also like normal variables. They refer to fixed values. They are alsocalled as literals.

Syntax:

const data_type variable_name; (or) const data_type *variable_name;

63)

Define volatile Keyword with Syntax?Ans:

When a variable is defined as volatile, the program may not change the value of thevariable explicitly.

Syntax:

volatile data_type variable_name; (or) volatile data_type *variable_name;

64)

What is a Macro?Ans:

Macros are the identifiers that represent statements or expressions. To associatemeaningful identifiers with constants, keywords, and statements or expressions.

65)

What is the difference between #include< > and #include" "?Ans

#include< >

Specifically used for built in header files.

#include" "

specifically used for used for user defined/created n header file

Define Array?Ans:

C Array is a collection of variables belongings to the same data type.OftAn Array is a collection of Homogeneous or similar data type elements having uniquevalues, and stored ata different locations. You can store group of data of same data typeIn an array.

69)

Types of an Array?Ans:

There are 2 types of C arrays. They are

1. One dimensional array

2) Multi dimensional array

70)

What are the Characteristics of Arrays?Ans:

An array holds elements that have the same data type.

Array elements are stored in subsequent memory locations

Two-dimensional array elements are stored row by row in subsequent memory locations.

Array name represents the address of the starting element

Define Pointer with Syntax and example?Ans:

C Pointer is a variable that stores/points the address of another variable. C Pointer isused to allocate memory dynamically i.e. at run time.

Syntax:

data_type *var_name;

Example :

int *p; char *p;

79)

What are the uses of Pointers?Ans:

Pointer is used in the following cases

It is used to access array elements.

It is used for dynamic memory allocation.

It is used in Call by reference.

It is used in data structures like trees, graph, linked list etc.

80)

What is the invalid pointer Arithmetic?Ans:

adding ,multiplying and dividing two pointers.

Shifting or masking pointer.

Addition of float or double to pointer

Assignment of a pointer of one type to a pointer of another type

81)

What is a pointer value and address?Ans:

A pointer value is a data object that refers to a memory location. Each memorylocation is numbered in the memory. The number attached to a memory location is calledthe address of the location.